

Actual trial process in expectation

Q1: How many cases among n -enrolled?

• Inputs: n -enrolled, n -fast, n -slow

• Key rates:

μ_f = annual rate of progression to disease (once infected) for fast progressors.

$\mu_s = "$ " for slow progressors.

μ_r = annual rate of reverting (once infected).

• In terms of random variables,

let X be the prob of progressing to disease (once infected) ~~by~~ ^{at} some specified time, for fast progressors.

Let Y be the same for slow progressors.

And let Z be the prob of reverting (once infected) ~~by~~ ^{at} some specified time.

We assume that:

$$\begin{cases} X \sim \text{Exponential}(\mu_f), \\ Y \sim \text{Exponential}(\mu_s), \\ Z \sim \text{Exponential}(\mu_r). \end{cases}$$

(time to event $\equiv$ Exponential distribution)

• Without reversion, the probability of becoming a case during the trial is:

$$P(\text{becoming a case}) = \begin{cases} P(X \leq 3) & \leftarrow \text{IF FAST,} \\ P(Y \leq 3) & \leftarrow \text{IF SLOW.} \end{cases}$$

• Now with reversion, we have an exponential race - there are two exponentially-distributed r.v.'s (μ_f/μ_s and μ_r) and we want to know what happens first (progressing or reverting).

~~we~~ Want $P(\text{progress} + \text{don't revert})$ during the trial.

• $P(\text{something happens}) = P(\text{progress OR revert} \text{ OR both})$

~~$P(X \leq 3 \text{ OR } Z \leq 3)$~~

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can't happen - assuming progression and reversion are mutually-exclusive

~~$P(X \leq 3 \text{ OR } Z \leq 3)$~~

→ It's not quite this because X and Z are events firing at the same time

Instead, $P(\text{something happens during trial}) = P(\min(X, Z) \leq 3)$.

μ_f for fast, μ_s for slow

• Claim $\min(X, Z) \sim \text{Exp}(\mu_f + \mu_r)$.

Proof

CDF: $P(\min(X, Z) \leq t) = 1 - P(\min(X, Z) > t)$

$$= 1 - P(X > t, Z > t)$$

$$= 1 - P(X > t) P(Z > t) \quad \text{as } X \text{ and } Z \text{ are independent}$$

$$= 1 - e^{-\mu_f t} e^{-\mu_r t} \quad \text{as } X \text{ and } Z \text{ are exponentially distributed}$$

$$= 1 - e^{-(\mu_f + \mu_r)t}$$

Hence $\min(X, Z)$ has CDF of ^{an} exponentially distributed r.v. with mean $(\mu_f + \mu_r)$.

The CDF exactly and uniquely defines a r.v.'s probability distribution.

Hence we have that: $\square$

$$\min(X, Z) \sim \text{Exp}(\mu_f + \mu_r). \quad \square$$

• So $P(\text{something happens during the trial}) = P(\min(X, Z) \leq 3)$

[or equivalently swap Y for X]

$$= \begin{cases} 1 - e^{-3(\mu_f + \mu_r)} & \leftarrow \text{IF FAST,} \\ 1 - e^{-3(\mu_s + \mu_r)} & \leftarrow \text{IF SLOW.} \end{cases}$$

• Lastly, what is the something? Is it a reversion or progression?

Claim: Assume an event has occurred. Then $P(\text{progression wins}) = \frac{\mu_f}{\mu_f + \mu_r}$.

Proof:

~~if the event is a reversion~~

[or μ_s if slow]

$$\begin{aligned} P(\text{progression wins}) &= P(X < Z) \quad (\text{or equivalently } Y \text{ if slow}) \\ &= \int_0^\infty P(X < Z | Z=z) f_Z(z) dz \quad \text{integrating over the joint pdf} \\ &\quad \text{(i.e. } X < Z \text{ for all values } z) \\ &= \int_0^\infty \int_0^z f_X(x) f_Z(z) dx dz \\ &= \int_0^\infty \int_0^z (\mu_f e^{-\mu_f x}) (\mu_r e^{-\mu_r z}) dx dz = \int_0^\infty \left[-\mu_r e^{-\mu_r z} e^{-\mu_f x} \right]_{x=0}^{x=z} dz \\ &= \int_0^\infty (\mu_r e^{-(\mu_f + \mu_r)z} - \mu_r e^{-\mu_r z}) dz = \left[\frac{\mu_r}{\mu_f + \mu_r} e^{-(\mu_f + \mu_r)z} - e^{-\mu_r z} \right]_{z=0}^{z=\infty} \\ &= 0 - \left(\frac{\mu_r}{\mu_f + \mu_r} - 1 \right) = 1 - \frac{\mu_r}{\mu_f + \mu_r} = \frac{\mu_f}{\mu_f + \mu_r}. \quad \square \end{aligned}$$

• Putting everything together, for each individual,
 $P(\text{becoming a case during trial}) = P(\text{something happens}) \times P(\text{the something is a progression})$

$$= \begin{cases} (1 - e^{-3(\mu_f + \mu_r)}) \frac{\mu_f}{(\mu_f + \mu_r)} & \text{IF FAST,} \\ (1 - e^{-3(\mu_s + \mu_r)}) \frac{\mu_s}{(\mu_s + \mu_r)} & \text{IF SLOW.} \end{cases}$$

Then multiply by n -fast / n -slow respectively.